TEXORA · LECTURE SHEET

Estimation

Parameters and statistics, the properties of a good point estimator, standard error, interval estimation and the correct meaning of a confidence level, confidence intervals for a mean and a proportion, and how to choose the sample size.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 7 · LEVEL 2 TERM 1

1. Parameters and statistics: the logic of inference

Statistical inference runs in the opposite direction to probability. Probability starts from a known population and predicts the behaviour of samples; inference starts from one observed sample and reasons back to the unknown population. **Estimation** is the part of inference concerned with the numerical values of population characteristics.

A **parameter** is a fixed but usually unknown number describing the population: the population mean μ, the population standard deviation σ, or the population proportion p of defective items. A **statistic** is a number computed from the sample: the sample mean $\bar{x}$, the sample standard deviation s, or the sample proportion. A statistic used to estimate a parameter is called an **estimator**, and the particular value it takes in one sample is an **estimate**.

CONCEPT	POPULATION (PARAMETER)	SAMPLE (STATISTIC)
Mean	μ, unknown and fixed	$\bar{x}$, computed and varies from sample to sample
Standard deviation	σ, unknown and fixed	s, computed from the sample
Proportion defective	p, unknown and fixed	Sample proportion, computed from the sample

Because a statistic varies from sample to sample, it has a distribution of its own, the **sampling distribution**. Everything in this topic, standard errors, confidence intervals and sample size formulae, is a statement about that sampling distribution. The central fact is that for large samples the sampling distribution of $\bar{x}$ is approximately normal, centred at μ, with a spread that shrinks as the sample size grows. Estimation is therefore possible: a single sample mean is unlikely to fall far from μ, and we can say how far is too far.

2. Point estimation

A **point estimate** is a single number offered as the value of a parameter. The natural candidates are the matching sample statistics: $\bar{x}$ estimates μ, s estimates σ, and the sample proportion estimates p. In the worked example carried through this sheet, 64 yarn samples give a mean mass of $\bar{x} = 52$ g, so the point estimate of the population mean μ is simply 52 g; in the proportion example, a sample proportion of 0.20 is the point estimate of the population defective proportion p.

A point estimate alone is incomplete, because it carries no statement of its own precision. Two samples could both give $\bar{x} = 52$ g, one from 4 specimens and one from 400, and they plainly deserve different trust. Point estimation theory therefore judges estimators by properties, and interval estimation, in Section 4, attaches the missing precision statement.

3. Properties of a good estimator

Four properties are traditionally required of a good point estimator. They should be learned as definitions with meanings, not as a bare list.

PROPERTY	MEANING
Unbiasedness	The estimator is unbiased if its expected value over all possible samples equals the parameter: E(estimator) = parameter. It hits the target on average, with no systematic overstatement or understatement. The sample mean $\bar{x}$ is unbiased for μ.
Consistency	The estimator is consistent if it closes in on the parameter as the sample size grows: its sampling distribution concentrates ever more tightly around the true value. Larger samples should mean better estimates, and consistency is the property that guarantees it.
Efficiency	Among unbiased estimators, the more efficient one has the smaller variance, that is, the tighter sampling distribution. An efficient estimator extracts more information from the same sample.
Sufficiency	An estimator is sufficient if it uses all the information in the sample about the parameter, so that no other statistic computed from the same sample can add anything. The sample mean is sufficient for μ when sampling from a normal population.

No single estimator is automatically best on every property for every parameter, which is why the properties are stated separately. In practice the sample mean scores well on all four for estimating a population mean, which justifies its central place in this course.

4. Standard error

The **standard error** (SE) is the standard deviation of a statistic sampling distribution. It measures the typical distance between the statistic and the parameter it estimates, and it is the unit in which all confidence interval widths are expressed. For the sample mean,

$$SE(\bar{x}) = \sigma / \sqrt{n}$$

where σ is the population standard deviation and n is the sample size. When σ is unknown, the sample standard deviation s is used in its place, which is standard practice for large samples. Two lessons are built into the formula. First, precision improves with sample size, but only as √n: quadrupling the sample only halves the standard error. Second, a more variable population is harder to estimate, in direct proportion to σ.

> **WORKED EXAMPLE**
>
> A study takes n = 64 yarn samples with sample mean $\bar{x}$ = 52 g and standard deviation σ (or s) = 8 g.
> SE = 8 / √64 = 8 / 8 = **1.0 g**.
> Sample means from repeated samples of 64 would typically fall within about 1 g of the true mean, with the normal pattern of variation described next.

5. Interval estimation and the meaning of confidence

An **interval estimate** gives a range of plausible values for the parameter, together with a **confidence level** such as 95% or 99% that states how reliable the method is. For a mean, when σ is known or the sample is large, the interval takes the form estimate plus or minus a margin:

$$\bar{x} \pm z \cdot \sigma / \sqrt{n} \quad \text{that is} \quad \bar{x} \pm z \cdot SE$$

where z is the standard normal value cutting off the required tail areas: z = 1.96 for 95% confidence and z = 2.58 for 99% confidence. The product z · SE is the **margin of error**. Higher confidence demands a larger z and therefore a wider interval: certainty is bought with width.

The interpretation must be stated carefully, because it is the most commonly mishandled idea in elementary statistics. A 95% confidence interval does **not** mean there is a 95% probability that this particular interval contains μ; μ is fixed, and any one computed interval either contains it or does not. The correct statement is about the **procedure**: if the sampling and interval construction were repeated many times, about 95% of the intervals produced would contain the true μ. We say we are 95% confident in the interval because we used a procedure that succeeds 95% of the time. Examiners look for exactly this distinction.

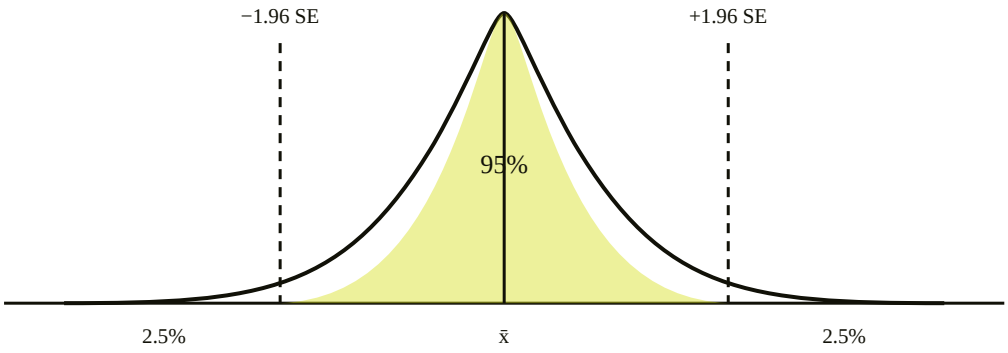


Figure 1 — Sampling distribution of the mean: the central 95% lies within ±1.96 standard errors of the centre

6. Confidence intervals for a mean: worked results

WORKED EXAMPLES

Data: n = 64, $\bar{x}$ = 52 g, σ (or s) = 8 g, so SE = 1.0 g.

95% CI = 52 ± 1.96 × 1.0 = 52 ± 1.96 → (**50.04, 53.96**) g.

99% CI = 52 ± 2.58 × 1.0 = 52 ± 2.58 → (**49.42, 54.58**) g.

The 99% interval is wider, as it must be: a more reliable procedure casts a wider net.

Reading the 95% result aloud is good practice: based on 64 samples, the mean yarn mass is estimated to lie between 50.04 g and 53.96 g, by a procedure that captures the true mean in 95% of repeated applications. Note what the interval does for a textile decision: if a buyer specification requires a mean of at least 50 g, the whole 95% interval lies above 50 g, which is reassuring evidence; had the interval straddled 50 g, the honest conclusion would be that the data cannot settle the question at this sample size.

7. Determining the sample size

Estimation should be planned, not left to habit. If the margin of error must not exceed a target value E at a stated confidence level, the required sample size follows by solving $E = z\,\sigma / \sqrt{n}$ for n:

$$n = (z\,\sigma / E)^2$$

The result is always rounded **up** to the next whole number, because rounding down would give a margin slightly larger than the target. A value of σ must be available before sampling, from a pilot study, earlier production data, or experience; without it, no honest sample size can be quoted.

WORKED EXAMPLE

Required: margin E = 1 g at 95% confidence, with σ = 8 g.

$n = (1.96 \times 8 / 1)^2 = (15.68)^2 = 245.86 \rightarrow$ round up to **246** samples.

Compare with the study above: n = 64 gave a margin of 1.96 g, so roughly halving the margin needs roughly four times the sample, exactly as the √n law predicts.

8. Confidence interval for a proportion

The same pattern applies to a proportion. If a sample of size n gives sample proportion $\hat{p}$ (written simply as p in the data below), its standard error is

$$SE = \sqrt{p\,(1 - p) / n}\,, \quad CI: p \pm z \cdot SE$$

WORKED EXAMPLE

A sample of n = 100 items gives a defective proportion p = 0.20.

$SE = \sqrt{(0.2 \times 0.8 / 100)} = \sqrt{0.0016} =$ **0.04**.

95% CI $= 0.20 \pm 1.96 \times 0.04 = 0.20 \pm 0.0784 \rightarrow$ (**0.122, 0.278**).

In words: the true defective proportion is estimated to lie between about 12.2% and 27.8% by the 95% procedure. The width shows how coarse a sample of 100 is for proportion work.

Point and interval estimation together give the complete answer to any estimation question: the point estimate states the best single value, 52 g for the mean or 0.20 for the proportion, and the interval states how much trust that value deserves at the chosen confidence level. Reporting one without the other is only half a result.

Glossary

TERM	MEANING
Parameter	A fixed, usually unknown number describing a population, such as μ, σ or p.
Statistic	A number computed from a sample, such as $\bar{x}$ or s; it varies from sample to sample.
Point estimate	A single number used as the estimated value of a parameter.
Unbiased estimator	An estimator whose expected value over all samples equals the parameter.
Standard error	The standard deviation of a statistic sampling distribution; for a mean, σ / √n.

Confidence interval	An interval of plausible parameter values built by a procedure with a stated success rate.
Confidence level	The long run proportion of intervals from the procedure that contain the true parameter, such as 95%.
Margin of error	The half width of a confidence interval, $z \cdot SE$.

Key points & summary

- A parameter describes the population and is fixed; a statistic describes the sample and varies, and that variation is governed by the sampling distribution.
- Good point estimators are unbiased, consistent, efficient and sufficient; the sample mean satisfies all four for estimating μ.
- Standard error of the mean is $\sigma / \sqrt{n}$; with $\sigma = 8$ and $n = 64$, SE = 1.0 g.
- The 95% CI for a mean is $\bar{x} \pm 1.96$ SE: here (50.04, 53.96) g; the 99% CI uses 2.58 and is wider: (49.42, 54.58) g.
- 95% confidence describes the procedure: about 95% of intervals built this way contain the true parameter; it is not a probability about one fixed interval.
- Sample size for margin E is $n = (z\sigma / E)^2$, always rounded up: E = 1 g at 95% with $\sigma = 8$ needs $n = 246$.
- For a proportion, $SE = \sqrt{p(1 - p)/n}$; $p = 0.20$ with $n = 100$ gives a 95% CI of (0.122, 0.278).

Practice

Answer in your own words; brief model answers follow.

1. Distinguish a parameter from a statistic, giving one textile example of each.
2. State the four properties of a good point estimator and explain unbiasedness in your own words.
3. For $n = 64$, $\bar{x} = 52$ g and $\sigma = 8$ g, compute the standard error and the 95% confidence interval for the mean.
4. For the same data, compute the 99% confidence interval and explain why it is wider than the 95% interval.
5. Explain carefully what 95% confidence means, and why it is wrong to say there is a 95% probability that the true mean lies inside one computed interval.
6. How many samples are needed for a margin of error of 1 g at 95% confidence when $\sigma = 8$ g? Show the calculation and state the rounding rule.

Answer key

1. A parameter is a fixed population value, for example the true mean breaking strength μ of all yarn produced in a week; a statistic is computed from a sample, for example the mean $\bar{x}$ of 64 tested specimens. The statistic estimates the parameter.
2. Unbiasedness, consistency, efficiency and sufficiency. Unbiasedness means the estimator averages to the true parameter over all possible samples, E(estimator) = parameter, so it has no systematic error in either direction.
3. SE = 8 / √64 = 1.0 g. 95% CI = 52 ± 1.96 × 1.0 = 52 ± 1.96, giving (50.04, 53.96) g.
4. 99% CI = 52 ± 2.58 × 1.0 = (49.42, 54.58) g. It is wider because greater confidence requires a larger multiplier z, so the procedure casts a wider net to succeed 99% of the time instead of 95%.
5. 95% confidence is a property of the procedure: in repeated sampling, about 95% of the intervals constructed in this way would contain the fixed true mean. A single computed interval either contains μ or does not; probability applies to the method across repetitions, not to the fixed parameter in one interval.
6. $n = (1.96 \times 8 / 1)^2 = 245.86$, rounded up to 246 samples. Rounding is always upward, because rounding down would leave the margin slightly above the target of 1 g.

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